**1-to'plam.** *O'xshash hadlarni qo'shing (musbat koeffitsientlar).*

1. $3x + 5x =$
2. $7y + 2y + y =$
3. $4a + 3b + 2a =$
4. $6x + 3 + 2x =$
5. $5m + 4n + 3m + n =$
6. $2x + 3x + 4x =$
7. $8p + 5q + 2p + 3q =$
8. $x + x + x + x =$

**2-to'plam.** *O'xshash hadlarni qo'shing – ayirish ishtirok etadi.*

1. $7x - 3x =$
2. $5a - 8a =$
3. $4x - 7 + 2x =$
4. $-3y + 5y - 2y =$
5. $6m - 4n - 2m + 7n =$
6. $-5x + 3x - 8 =$
7. $2a - 5a + 6a =$
8. $9x - 4x - x =$

**3-to'plam.** *Taqsimot qonunini qo'llang (musbat ko'paytuvchi).*

1. $3(x+4) =$
2. $5(2x-3) =$
3. $2(3a+4b) =$
4. $4(x-1) =$
5. $6(2+3x) =$
6. $7(x+y) =$
7. $3(4x-2) =$
8. $2(5x+3) =$

**4-to'plam.** *Taqsimot qonuni – manfiy ko'paytuvchini har bir hadga tarqating.*

1. $-2(x+3) =$
2. $-(3x-5) =$
3. $-4(2x-1) =$
4. $-(x+7) =$
5. $-3(4-x) =$
6. $-(2x-3y) =$
7. $-5(x+2) =$
8. $-(-x+4) =$

**5-to'plam.** *Aralash – avval taqsimlang, so'ng o'xshash hadlarni qo'shing.*

1. $3(x+2)+4x=$

2. $5(2x-1)-3x=$

3. $2(x+3)-(x-1)=$

4. $4(x-2)+3(x+1)=$

5. $-(2x+3)+5x=$

6. $3(x-1)-2(x+2)=$

7. $2(3x+1)-(4x-5)=$

8. $5-2(x-3)=$

**6-to'plam.** *Xatoni toping – buzilgan qadamni ayting, to'g'ri soddalashtirilgan ko'rinishni bering.*

1. $-(3x-5)$ ifodasini soddalashtirishda ko'rsatilgan yechimda xato bor:

   "$-3x-5$"

   Xato va to'g'ri javob: .....

2. $4x+3x^2$ ifodasini soddalashtirishda ko'rsatilgan yechimda xato bor:

   "$7x^2$"

   Xato va to'g'ri javob: .....

3. $-2(x-4)$ ifodasini soddalashtirishda ko'rsatilgan yechimda xato bor:

   "$-2x-8$"

   Xato va to'g'ri javob: .....

4. $5x-(2x-3)$ ifodasini soddalashtirishda ko'rsatilgan yechimda xato bor:

   "$5x-2x-3=3x-3$"

   Xato va to'g'ri javob: .....

5. $3(2x+1)-4$ ifodasini soddalashtirishda ko'rsatilgan yechimda xato bor:

   "$6x+1-4=6x-3$"

   Xato va to'g'ri javob: .....

**7-to'plam.** *Bonus / qiyin – ichma-ich qavslar va ikki o'zgaruvchili ifodalar.*

1 $2\,[3(x+1)-2x]+5=$

2 $-3(x-2)+2(x+4)-5=$

3 $4x-2\,[x-(3x+1)]=$

4 $5(a-2b)-3(2a-b)=$

5 $-\,[2(x+3)-(x-1)]+4x=$

**8-to'plam.** *SAT'da qanday uchraydi – had, koeffitsient va teng kuchli ifodalar.*

1 $7x - 4y + 9$ ifodasida $y$ ning koeffitsienti nechaga teng?

(A) 4 (B) $-4$

(C) 7 (D) 9

2 $5x + 3 - 2x + 8$ ifodasiga teng ifodani toping.

(A) $3x + 11$ (B) $7x + 11$

(C) $3x + 5$ (D) $7x + 5$

3 $4(2x - 3)$ ifodasiga teng ifodani toping.

(A) $8x - 3$ (B) $6x - 12$

(C) $2x - 12$ (D) $8x - 12$

4 $-3(x - 5)$ ifodasiga teng ifodani toping.

(A) $-3x - 5$ (B) $-3x - 15$

(C) $-3x + 15$ (D) $3x - 15$

5 $6x + 2(x - 3)$ ifodasiga teng ifodani toping.

(A) $8x - 3$ (B) $6x + 1$

(C) $8x + 6$ (D) $8x - 6$

6 To'rtburchakning eni $x$, uzunligi esa enidan ikki baravar ortiq va yana 3 birlik ko'p. Perimetri uchun qaysi ifoda to'g'ri?

(A) $3x + 3$ (B) $6x + 6$

(C) $6x + 3$ (D) $3x + 6$

7 $2(3x + 4) - (x - 5)$ ifodasiga teng ifodani toping.

(A) $5x + 3$ (B) $5x - 13$

(C) $5x + 13$ (D) $7x + 13$

8 Do'konda daftar narxi \$3, xaridda bir martalik \$5 chegirma beriladi. $n$ ta daftar uchun jami narx (dollarda) qanday ifoda bilan beriladi?

(A) $3n + 5$ (B) $3(n - 5)$

(C) $5n - 3$ (D) $3n - 5$

9 $-2(3x - 1) + 4x$ ifodasiga teng ifodani toping.

(A) $2x + 2$ (B) $-2x + 2$

(C) $-2x - 2$ (D) $10x + 2$

10 $8 - 5x + 2x^2$ ifodasida erkin had nechaga teng?

(A) 8 (B) $-5x$

(C) $2x^2$ (D) $-5$

11 $3x + 7 - (x + 7)$ ifodasiga teng ifodani toping.

(A) $2x + 14$ (B) $4x$

(C) $2x - 14$ (D) $2x$

12 Telefon tarifi oyiga \$15 + har gigabayt uchun \$2. $g$ gigabayt uchun jami narx (dollarda) qanday ifoda bilan beriladi?

(A) $15g + 2$ (B) $2g - 15$

(C) $2g + 15$ (D) $17g$

# Javoblar

| To'plam | # | Javob | # | Javob |
|---|---|---|---|---|
| 1-to'plam | 1 | $8x$ | 2 | $10y$ |
| | 3 | $6a + 3b$ | 4 | $8x + 3$ |
| | 5 | $8m + 5n$ | 6 | $9x$ |
| | 7 | $10p + 8q$ | 8 | $4x$ |
| 2-to'plam | 1 | $4x$ | 2 | $-3a$ |
| | 3 | $6x - 7$ | 4 | $0$ |
| | 5 | $4m + 3n$ | 6 | $-2x - 8$ |
| | 7 | $3a$ | 8 | $4x$ |
| 3-to'plam | 1 | $3x + 12$ | 2 | $10x - 15$ |
| | 3 | $6a + 8b$ | 4 | $4x - 4$ |
| | 5 | $18x + 12$ | 6 | $7x + 7y$ |
| | 7 | $12x - 6$ | 8 | $10x + 6$ |
| 4-to'plam | 1 | $-2x - 6$ | 2 | $5 - 3x$ |
| | 3 | $4 - 8x$ | 4 | $-x - 7$ |
| | 5 | $3x - 12$ | 6 | $-2x + 3y$ |
| | 7 | $-5x - 10$ | 8 | $x - 4$ |
| 5-to'plam | 1 | $7x + 6$ | 2 | $7x - 5$ |
| | 3 | $x + 7$ | 4 | $7x - 5$ |
| | 5 | $3x - 3$ | 6 | $x - 7$ |
| | 7 | $2x + 7$ | 8 | $11 - 2x$ |
| 7-to'plam | 1 | $2x + 11$ | 2 | $9 - x$ |
| | 3 | $8x + 2$ | 4 | $-a - 7b$ |
| | 5 | $3x - 7$ | – | – |
| 8-to'plam | 1 | B | 2 | A |
| | 3 | D | 4 | C |
| | 5 | D | 6 | B |
| | 7 | C | 8 | D |
| | 9 | B | 10 | A |
| | 11 | D | 12 | C |
| 6-to'plam | 1. Xato – manfiy ishora faqat $3x$ ga tarqatilgan, $-5$ ga emas. To'g'ri: $-(3x - 5) = 5 - 3x$. | | | |
| | 2. Xato – $x$ va $x^2$ o'xshash had emas (daraja boshqa), qo'shib bo'lmaydi. To'g'ri: $3x^2 + 4x$ (allaqachon sodda). | | | |
| | 3. Xato – $-2 \times (-4) = +8$ bo'lishi kerak edi, ishora tushib qolgan. To'g'ri: $-2(x - 4) = 8 - 2x$. | | | |
| | 4. Xato – qavs oldidagi manfiy ishora faqat $2x$ ga tarqatilgan, $-3$ ga emas. To'g'ri: $5x - (2x - 3) = 3x + 3$. | | | |
| | 5. Xato – $3 \times 1$ hisoblanmagan (ikkinchi hadga tarqatish o'tkazib yuborilgan). To'g'ri: $3(2x + 1) - 4 = 6x - 1$. | | | |

Video yechimlar va qo'shimcha testlar uchun quyidagi QR kodlarni skanerlang:

QR kod

YouTube

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Telegram